Worksheet 1.4 — Transformations of Functions

Grade 12 Advanced Functions (MHF4U) · Unit 1: Polynomial Functions

The Big Ideas

Apply $g(x) = a\,f(k(x-d)) + c$ to the parent $f(x) = x^n$.

- a: vertical stretch/reflection. k: horizontal stretch/reflection.
- d: horizontal shift (opposite to the sign). c: vertical shift.
- Stretches/reflections first, then translations.

Core Concepts

A single framework

Any base function f transforms through $y = af(k(x-d)) + c$, where each constant stretches, reflects, or shifts the graph.

Vertical vs horizontal

a and c act vertically (stretch by a, shift by c); k and d act horizontally (compress by k, shift by d).

Order and mapping

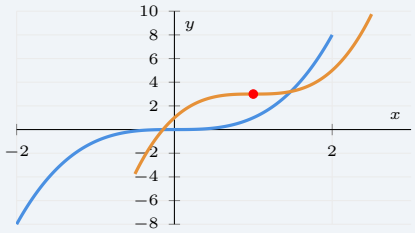
Apply horizontal changes to x and vertical changes to y: each point maps as $(x, y) \to \left(\frac{x}{k} + d,\ ay + c\right)$.

Worked Examples

Example 1 — Describe

Describe $y = 2(x-1)^3 + 3$ from $y = x^3$.

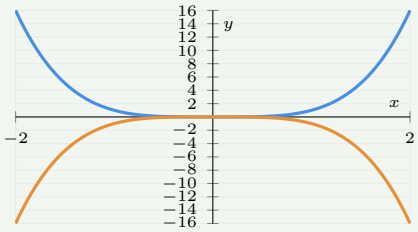
Solution. $a = 2$ stretch; $d = 1$ right; $c = 3$ up centred at $(1, 3)$:



Example 2 — Reflection

Describe $y = -x^4$.

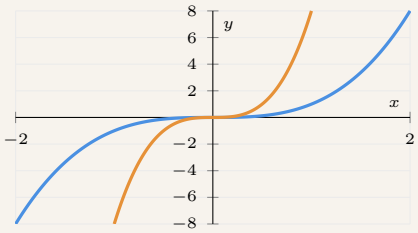
Solution. $a = -1$: reflection in the x-axis (opens down):



Example 3 — Horizontal stretch

What does $k = 2$ do in $y = (2x)^3$?

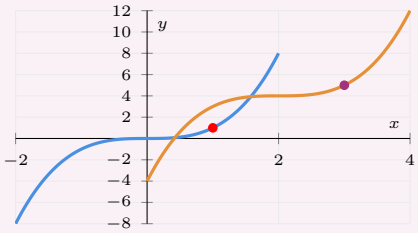
Solution. Horizontal compression by $\frac{1}{2}$ (squeezed toward the y-axis):



Example 4 — Map a point

$(1, 1)$ on $y = x^3$ maps where on $y = (x - 2)^3 + 4$?

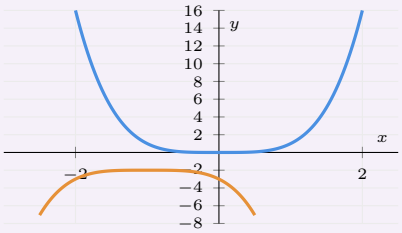
Solution. Right 2, up 4: $(3, 5)$:



Example 5 — Write the equation

Write $y = x^4$ reflected, left 1, down 2.

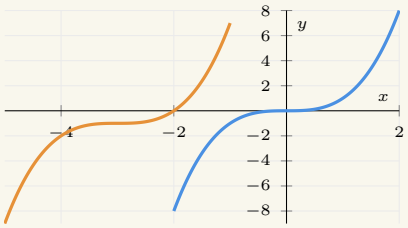
Solution. $a = -1,\ d = -1,\ c = -2$: $y = -(x+1)^4 - 2$:



Example 6 — Describe

Describe $y = (x+3)^3 - 1$.

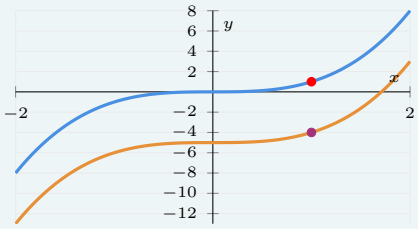
Solution. Left 3, down 1:



Example 7 — Map a point

$(1, 1)$ on $y = x^3$ maps where on $y = x^3 - 5$?

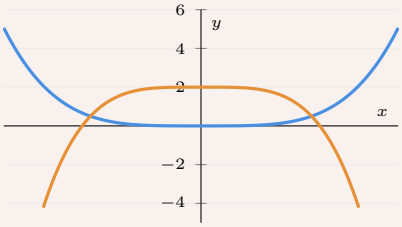
Solution. $(1, -4)$:



Example 8 — Stretch & reflect

Describe $y = -3x^4 + 2$.

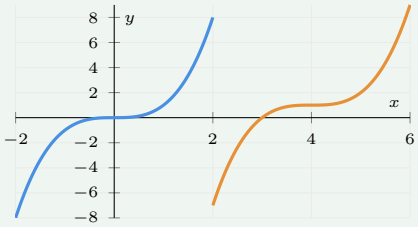
Solution. Reflect, stretch by 3, up 2:



Example 9 — Write the equation

Write $y = x^3$ shifted right 4 and up 1.

Solution. $y = (x-4)^3 + 1$:



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Describe $y = (x-6)^3 + 2$ from $y = x^3$.

Question 2

What does $a = -2$ do?

Question 3

Where does $(1, 1)$ on $y = x^3$ map under $y = x^3 + 7$?

Question 4

Write $y = x^3$ shifted left 3 and down 5.

Question 5

Describe $y = -4x^4 - 1$.

Question 6

Write $y = x^4$ reflected, right 2 and up 6.

Question 7

Where does $(1, 1)$ on $y = x^3$ map under $y = (x + 1)^3 - 3$?

Question 8

What does $k = 3$ do?

Question 9

Describe $y = 2(x + 1)^4$ from $y = x^4$.

Question 10

Write $y = x^3$ reflected in the x-axis.

Question 11

State the vertex-like centre of $y = (x - 5)^3 + 2$.

Question 12

Describe $y = \frac{1}{2}x^3$.

Question 13 — Challenge

Fully describe $y = -2(x + 1)^3 - 3$.

Question Recap & Answer Key

Question Recap

1. Describe $y = (x-6)^3 + 2$ from $y = x^3$.
2. What does $a = -2$ do?
3. Where does $(1, 1)$ on $y = x^3$ map under $y = x^3 + 7$?
4. Write $y = x^3$ shifted left 3 and down 5.
5. Describe $y = -4x^4 - 1$.
6. Write $y = x^4$ reflected, right 2 and up 6.
7. Where does $(1, 1)$ on $y = x^3$ map under $y = (x+1)^3 - 3$?
8. What does $k = 3$ do?
9. Describe $y = 2(x+1)^4$ from $y = x^4$.
10. Write $y = x^3$ reflected in the x-axis.
11. State the vertex-like centre of $y = (x-5)^3 + 2$.
12. Describe $y = \frac{1}{2}x^3$.
13. Fully describe $y = -2(x+1)^3 - 3$.

✓ Answer Key

1. right 6, up 2
2. reflect and stretch by 2
3. $(1, 8)$
4. $y = (x+3)^3 - 5$
5. reflect, stretch 4, down 1
6. $y = -(x-2)^4 + 6$
7. $(0, -2)$
8. horizontal compression by $\frac{1}{3}$
9. stretch 2, left 1
10. $y = -x^3$
11. $(5, 2)$
12. vertical compression by $\frac{1}{2}$
13. reflect, stretch 2, left 1, down 3